

What are the complications?

Last revised in March 2025

Possible complications of type 2 diabetes include:

- **Macrovascular**

- Atherosclerotic cardiovascular disease (CVD)
 - CVD including stroke and peripheral arterial disease is the leading cause of death in people with type 2 diabetes [[Davies, 2018 \(/topics/diabetes-type-2/references/\)](#)].
 - People with type 2 diabetes have a two-fold increased risk of stroke within the first 5 years of diagnosis compared with the general population [[Diabetes UK, 2019a \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topic on [Stroke and TIA \(/topics/stroke-tia/\)](#) for more information.
 - People with type 2 diabetes are 2.5 times more likely to have a myocardial infarction and develop heart failure than people without diabetes [[Diabetes UK, 2019c \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topics on [Angina \(/topics/angina/\)](#) and [Heart failure - chronic \(/topics/heart-failure-chronic/\)](#) for more information.
 - Peripheral arterial disease is a risk factor for the development of diabetic foot disease and foot ulcers and is present in up to 50% of people with diabetic foot ulcer [[Schaper, 2020 \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topic on [Peripheral arterial disease \(/topics/peripheral-arterial-disease/\)](#) for more information.

- **Microvascular**

- Diabetic kidney disease
 - This may be caused by diabetic nephropathy (hyperglycaemia-induced glomerular disease resulting in albuminuria in the absence of other causes of albuminuria), hypertension, and renal atheroma or ischaemia, which cause varying degrees of glomerulosclerosis and tubular fibrosis [[Mark, 2017 \(/topics/diabetes-type-2/references/\)](#); [Lewis, 2019 \(/topics/diabetes-type-2/references/\)](#)]. Proteinuria is usually the first sign of diabetic kidney disease and is a marker of increased cardiovascular risk [[Lewis, 2019 \(/topics/diabetes-type-2/references/\)](#)].
 - Diabetes is the commonest cause of chronic kidney disease (CKD) and end-stage renal disease (ESRD), which may require renal replacement therapy or kidney transplantation [[Winocour, 2020 \(/topics/diabetes-type-2/references/\)](#)]. Diabetes is the most commonly identified primary renal disease for people starting renal replacement therapy [[Renal Association, 2020 \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](#) for more information.
- Retinopathy
 - Diabetes is the leading cause of preventable sight loss in the UK, and 7% of people who are newly registered blind in England and Wales have lost their sight due to diabetes [[Diabetes UK, 2019c \(/topics/diabetes-type-2/references/\)](#)].
 - More than one-quarter of people with type 2 diabetes have some form of retinopathy [[Diabetes UK, 2019a \(/topics/diabetes-type-2/references/\)](#)].
 - Within 20 years of diagnosis, two-thirds of people with type 2 diabetes will have some degree of retinopathy [[Diabetes UK, 2019a \(/topics/diabetes-type-](#)

[2/references/](#)).

- Peripheral neuropathy
 - This is estimated to affect up to 50% of people with diabetes, and commonly presents as distal and symmetrical polyneuropathy [[Guttormsen, 2017 \(/topics/diabetes-type-2/references/\)](#)].
 - Painful neuropathy may cause symptoms of numbness, burning or shooting pain, tingling and/or paraesthesia of the hands and/or feet typically in a stocking and glove distribution, often at night. Over time it can progress to persistent neuropathic pain [[Donoghue, 2018 \(/topics/diabetes-type-2/references/\)](#); [Baker, 2020 \(/topics/diabetes-type-2/references/\)](#)].
- Autonomic neuropathy
 - This can affect multiple systems and includes postural hypotension, gastroparesis (delayed gastric emptying), unexplained diarrhoea, inadequate bladder emptying, sexual dysfunction including erectile dysfunction, sweating abnormalities, and impaired awareness of hypoglycaemia [[Donoghue, 2018 \(/topics/diabetes-type-2/references/\)](#); [NICE, 2020a \(/topics/diabetes-type-2/references/\)](#)].
- **Foot problems**
 - Diabetic foot problems may result from a combination of peripheral arterial disease and peripheral neuropathy, due to loss of protective foot sensation and possible deformity leading to abnormal loading of the foot. Minor trauma, for example from ill-fitting shoes or injury can precipitate foot ulceration [[Baker, 2020 \(/topics/diabetes-type-2/references/\)](#); [Schaper, 2020 \(/topics/diabetes-type-2/references/\)](#)].
 - Limb- or life-threatening complications of foot disease include foot ulceration with signs of sepsis, lower limb ischaemia risking limb amputation, deep-seated soft tissue or bone infection (osteomyelitis), and gangrene [[NICE, 2019b \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topics on [Sepsis \(/topics/sepsis/\)](#) and [Peripheral arterial disease \(/topics/peripheral-arterial-disease/\)](#) for more information.
- **Metabolic**
 - Dyslipidaemia
 - Type 2 diabetes is commonly associated with dyslipidaemia, which is a risk factor for CVD [[NICE, 2020a \(/topics/diabetes-type-2/references/\)](#)]. See the CKS topics on [Lipid modification - CVD prevention \(/topics/lipid-modification-cvd-prevention/\)](#) and [CVD risk assessment and management \(/topics/cvd-risk-assessment-management/\)](#) for more information.
 - Diabetic ketoacidosis (DKA)
 - This is the metabolic state characterized by the triad of hyperglycaemia, metabolic acidosis, and ketonaemia. It is a life-threatening emergency that may lead to dehydration and electrolyte imbalances, myocardial infarction, shock, cerebral oedema, and aspiration pneumonia. Although more common in type 1 diabetes, people with type 2 diabetes can also develop DKA [[French, 2019 \(/topics/diabetes-type-2/references/\)](#); [BSPED, 2020 \(/topics/diabetes-type-2/references/\)](#)].
 - People with a history of recurrent DKA are at high risk for all-cause mortality [[French, 2019 \(/topics/diabetes-type-2/references/\)](#)].
 - Hyperosmolar hyperglycaemic state (HHS)
 - This is a rare but potentially life-threatening emergency that occurs more typically in older people with type 2 diabetes and other co-morbidities, who develop severe hyperglycaemia, often over several weeks, often due to a combination of acute illness and dehydration. It may present with urinary frequency, thirst, nausea, confusion, and severe dehydration [[Down, 2020 \(/topics/diabetes-type-2/references/\)](#)].

- **Psychosocial**
 - Diabetes may negatively impact on quality of life and is associated with an increased risk of anxiety and depression. The prevalence of depression is about twice as high in people with diabetes as in the general population [Milne, 2020 (</topics/diabetes-type-2/references/>)]. It may be exacerbated by other factors such as chronic painful peripheral neuropathy [Guttormsen, 2017 (</topics/diabetes-type-2/references/>)]. See the CKS topics on [Generalized anxiety disorder](/topics/generalized-anxiety-disorder/) (</topics/generalized-anxiety-disorder/>) and [Depression](/topics/depression/) (</topics/depression/>) for more information.
 - Children and young people may experience depression, behavioural problems, family/relationship difficulties, and increased risk-taking behaviour (including non-adherence to treatment) [Candler, 2018a (</topics/diabetes-type-2/references/>); Zeitler, 2018 (</topics/diabetes-type-2/references/>)]. See the CKS topics on [Depression in children](/topics/depression-in-children/) (</topics/depression-in-children/>) and [Conduct disorders in children and young people](/topics/conduct-disorders-in-children-young-people/) (</topics/conduct-disorders-in-children-young-people/>) for more information.
- **Reduced life expectancy**
 - It is estimated that diabetes accounts for about 15–16% of deaths in England, with life expectancy for people with type 2 diabetes reduced by an average of up to 10 years. Diabetes UK estimates that in people with type 2 diabetes, the risk of death is 32% greater than expected, and is mostly in younger people and females [Diabetes UK, 2015 (</topics/diabetes-type-2/references/>)].

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